blacklistManagement HTTP/TLS/JSON

Interface Design Description

Service ID: *"blacklistManagement"*

Abstract
This document describes the HTTP/1.1 protocol with TLS payload security and JSON payload encoding variant of the `blacklistManagement` service of the Blacklist system.

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1 Overview

This document describes the `blacklistManagement` service interface of the Blacklist system, which provides operator management of the blacklist: adding systems with a reason and expiry, querying entries and inactivating them.

It's implemented using protocol, encoding as stated in below table:

Profile type	Type	Version
Transfer protocol	HTTP	1.1 [1]
Data encryption	TLS	1.3 [2], 1.2 [3]
Encoding	JSON	RFC 8259 [4]
Compression	N/A	-
Semantics	N/A	-
Ontology	N/A	-

Table 1: Communication and semantics details used for the `blacklistManagement` service interface

This document provides the Interface Design Description IDD to the *blacklistManagement – Service Description* document [5]. For further details about how this service is meant to be used, please consult that document.

The rest of this document describes how to realize the `blacklistManagement` service interface in detail. Both in terms of its operations (Section 2) and its information model (Section 3).

1.1 Interface registration

The service is registered in the ServiceRegistry with the interface template `generic_https`, policy `CERT_AUTH`, access address `blacklist`, port 8464 and base path `/blacklist/mgmt`. The `operations` property of the interface lists the operations below with their method and path relative to the base path (path parameters are placeholders such as `{id}` that the consumer fills in):

Operation	Method	Path
`query`	POST	`/query`
`create`	POST	`/create`
`remove`	DELETE	`/remove`

2 Service Operations

The interfaces of the `blacklistManagement` service, its operations, data models and implementation are provided below. In particular, each subsection first names the HTTP method and path used to call the operation, after which it names the abstract operation from the `blacklistManagement` SD document, as well as input and output types.

All operations require the requester's identity, presented according to the provider's authentication policy: the client certificate (mutual TLS) and, with the `outsourced` policy, `Authorization: Bearer IDENTITY-TOKEN//<token>`; systems holding the Authentication secret key send `Bearer AUTHENTICATOR-KEY//<system>//<hash>`, and with the `declared` policy `Bearer SYSTEM//<system>` is sent. Request and response bodies are `application/json` unless stated otherwise. Errors are answered with an ErrorMessage body and the status codes `400 Bad Request` (invalid parameter), `401 Unauthorized` (missing or invalid identity), `403 Forbidden` (not permitted, e.g. by the management, authorization or blacklist filter), `404 Not Found`, `408 Request Timeout`, `423 Locked`, `500 Internal Server Error` and `503 Service Unavailable` (a system it depends on failed).

2.1 POST /blacklist/mgmt/query

Operation: query
Input: BlacklistQueryRequest
Output: BlacklistEntryListResponse

Returns a page of entries matching the filter.

A successful call returns `200 OK` with the output in the body.

```
POST /blacklist/mgmt/query HTTP/1.1
Host: blacklist:8464
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "pagination": {
    "page": 0,
    "size": 10,
    "direction": "ASC",
    "sortField": "createdAt"
  },
  "systemNames": [
    "TemperatureSensor"
  ],
  "mode": "ACTIVES",
  "issuers": [
    "string"
  ],
  "revokers": [
    "string"
  ],
  "reason": "Compromised certificate",
  "alivesAt": "2026-10-22T12:00:00Z"
}
```

Listing 1: An query invocation.

```
{
  "entries": [
    {
      "systemName": "TemperatureSensor",
      "createdBy": "Sysop",
      "revokedBy": "string",
      "createdAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z",
```

```
      "reason": "Compromised certificate",
      "expiresAt": "2026-10-22T12:00:00Z",
      "active": true
    }
  ],
  "count": 1
}
```

Listing 2: An query response.

2.2 POST /blacklist/mgmt/create

Operation: create
Input: BlacklistCreateListRequest
Output: BlacklistEntryListResponse

Adds blacklist entries with reason and optional expiry.
A successful call returns `200 OK` with the output in the body.

```
POST /blacklist/mgmt/create HTTP/1.1
Host: blacklist:8464
Authorization: Bearer IDENTITY-TOKEN//<token>
Content-Type: application/json

{
  "entities": [
    {
      "systemName": "TemperatureSensor",
      "expiresAt": "2026-10-22T12:00:00Z",
      "reason": "Compromised certificate"
    }
  ]
}
```

Listing 3: An create invocation.

```
{
  "entries": [
    {
      "systemName": "TemperatureSensor",
      "createdBy": "Sysop",
      "revokedBy": "string",
      "createdAt": "2026-10-22T12:00:00Z",
      "updatedAt": "2026-10-22T12:00:00Z",
      "reason": "Compromised certificate",
      "expiresAt": "2026-10-22T12:00:00Z",
      "active": true
    }
  ],
  "count": 1
}
```

Listing 4: An create response.

2.3 DELETE /blacklist/mgmt/remove?names=*{names}*

Operation: remove
Input: Array<String>

Inactivates the entries of the named systems.
Query parameters: *names* (List<String>, required). List values are repeated or comma-separated.

A successful call returns `200 OK`.

```
DELETE /blacklist/mgmt/remove?names=<names> HTTP/1.1
Host: blacklist:8464
Authorization: Bearer IDENTITY-TOKEN//<token>
```

Listing 5: An remove invocation.

3 Data Models

Here, all data objects that can be part of the service calls associated with this service are listed in alphabetic order. Note that each subsection, which describes one type of object, begins with the *struct* keyword, which is meant to denote a JSON Object that must contain certain fields, or names, with values conforming to explicitly named types. Fields that are not needed may be omitted or `null` in requests; empty fields are omitted in responses. As a complement to the primary types defined in this section, there is also a list of secondary types in Section 3.8, which are used to represent things like hashes, identifiers and texts.

3.1 struct BlacklistCreateListRequest

Entries to add to the blacklist.

Object Field	Value Type	Description
"entities"	Array<BlacklistCreateRequest>	Entries to create.

3.2 struct BlacklistCreateRequest

One system to blacklist.

Object Field	Value Type	Description
"systemName"	String	System to blacklist.
"expiresAt"	DateTime	UTC expiry time; no value means no expiry.
"reason"	String	Reason of the blacklisting.

3.3 struct BlacklistEntry

A blacklist entry.

Object Field	Value Type	Description
"systemName"	String	The blacklisted system.
"createdBy"	String	System that created the entry.
"revokedBy"	String	System that inactivated the entry, if any.
"createdAt"	DateTime	UTC time of creation.
"updatedAt"	DateTime	UTC time of the last change.
"reason"	String	Reason of the blacklisting.
"expiresAt"	DateTime	Expiry of the entry; no value means permanent.
"active"	Boolean	Whether the entry is in force.

3.4 struct BlacklistEntryListResponse

List of blacklist entries.

Object Field	Value Type	Description
"entries"	Array<BlacklistEntry>	Matching records of the current page.
"count"	Number	Total number of matching records (all pages).

3.5 struct BlacklistQueryRequest

Paged filter for blacklist entries.

Object Field	Value Type	Description
"pagination"	Page	Paging and sorting of the result.
"systemNames"	Array<String>	System names to match.
"mode"	String	Which entries to return: `ALL`, `ACTIVES` or `INACTIVES`.
"issuers"	Array<String>	Systems that created the entries.
"revokers"	Array<String>	Systems that inactivated the entries.
"reason"	String	Reason of the blacklisting.
"alivesAt"	DateTime	Only entries in force at this time.

3.6 struct ErrorMessage

Object Field	Value Type	Description
"errorMessage"	String	Error text, if the event is an error.
"errorCode"	Number	
"exceptionType"	ExceptionType	
"origin"	String	

3.7 struct Page

Paging and sorting; all fields optional.

Object Field	Value Type	Description
"page"	Number	Zero-based page number.
"size"	Number	Number of records per page (at most `max.page.size`, default 1000).
"direction"	String	Sort direction, `ASC` or `DESC`.
"sortField"	String	Field to sort by.

3.8 Primitives

As all messages are encoded using the JSON format [4], the following primitive constructs, part of that standard, become available. Note that the official standard is defined in terms of parsing rules, while this list only concerns syntactic information. Furthermore, the Object and Array types are given optional generic type parameters, which are used in this document to signify when pair values or elements are expected to conform to certain types.

Type	Description
Array	A JSON Array [4]; Array<A> has elements of type A.
Boolean	JSON Boolean, `true` or `false`.
DateTime	JSON String with an ISO 8601 UTC timestamp, e.g. `2026-10-22T12:00:00Z`.
ExceptionType	One of `ARROWHEAD`, `INVALID_PARAMETER`, `AUTH`, `FORBIDDEN`, `DATA_NOT_FOUND`, `TIMEOUT`, `LOCKED`, `INTERNAL_SERVER_ERROR`, `EXTERNAL_SERVER_ERROR` (JSON String).
Number	JSON Number, an integer.
Object	JSON Object; Object<A> has string keys and values of type A.
String	JSON String.

4 References

[1] R. Fielding, M. Nottingham, and J. Reschke, “Http/1.1,” IETF, RFC 9112, 2022.

[2] E. Rescorla, “The transport layer security (tls) protocol version 1.3,” IETF, RFC 8446, 2018.

[3] T. Dierks and E. Rescorla, “The transport layer security (tls) protocol version 1.2,” IETF, RFC 5246, 2008.

[4] T. Bray, “The javascript object notation (json) data interchange format,” IETF, RFC 8259, 2017.

[5] J. Delsing, “blacklistManagement – Service Description (Blacklist),” Eclipse Arrowhead, SD 5.2.2, 2026, blacklist-v5.2.2-blacklistManagement-SD.pdf.

5 Revision History

5.1 Amendments

No.	Date	Version	Subject of Amendments	Author
1	2026-10-22	5.2.2	Document created from the Arrowhead LaTeX templates for the 5.2.2 Java Spring implementation, including the local extensions of this workspace.	Jerker Delsing
2	2026-10-30	5.2.2	Checked against the primary Java source tree (Eclipse-Arrowhead-java): services, operations, paths and data models unchanged; missing primitive definitions added.	Claude Opus 5.5 (draft)

5.2 Quality Assurance

No.	Date	Version	Approved by
1		5.2.2	